



The University of Edinburgh

College of Science and Engineering
School of Mathematics

Mathematics 3 Honours
MATH10003 Financial Mathematics

Monday 17th December 2018

2.30pm–4.30pm

Chairman of Examiners – Dr A Olde Daalhuis
External Examiner – Professor J McColl

Attempt **ALL** questions

Calculators and other electronic aids:

A scientific calculator is permitted in this examination.

It must not be a graphical calculator.

It must not be able to communicate with any other devices.

This examination will be marked anonymously.

(1) Denote by $(W_t)_{t \geq 0}$ the standard Brownian motion on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

(a) Define the stochastic process $(Q_t)_{t \in [0,1]}$ as $Q_t := W_t - tW_1$.

(i) Show that $\mathbb{E}[Q_t] = 0$ and $\mathbb{E}[Q_t Q_s] = \min\{t, s\} - ts$ for any $t, s \in [0, 1]$.

(ii) Is Q a Brownian motion? Justify briefly your answer.

[5 marks]

(b) Consider the stochastic differential equation

$$dX_t = -\mu X_t dt + \sigma dW_t, \quad t \geq 0,$$

with a given deterministic initial value x_0 , where $\mu > 0$ and $\sigma \neq 0$ are some real constants.

(i) Show that the solution X of the stochastic differential equation above is given by

$$X_t = x_0 e^{-\mu t} + \sigma \int_0^t e^{-\mu(t-s)} dW_s, \quad t \geq 0.$$

[4 marks]

(ii) The spot rate for the Vasicek model for interest rates satisfies the stochastic differential equation (μ, σ, a, c are some real strictly positive constants)

$$dr_t = \mu(a - r_t)dt + \sigma dW_t, \quad t \geq 0, \quad r_0 = c.$$

Possibly with the help of the process $X_t := r_t - a$ (for $t \geq 0$) and its SDE, show that

$$\lim_{t \rightarrow +\infty} \mathbb{E}[|r_t - a|^2] = \frac{\sigma^2}{2\mu}.$$

[7 marks]

(c) Let $T > 0$. Take $g : \mathbb{R} \rightarrow \mathbb{R}$ to be a differentiable function and denote $g'(x) = \frac{d}{dx}g(x)$. Let $(X_t)_{t \in [0, T]}$ be the solution to the stochastic differential equation

$$X_t = x_0 + \int_0^t \frac{1}{2} g(X_s) g'(X_s) ds + \int_0^t g(X_s) dW_s, \quad t \in [0, T], \quad x \in \mathbb{R},$$

Given the function g , define the additional function $h : \mathbb{R} \rightarrow \mathbb{R}$ as

$$h(x) := \int_0^x \frac{1}{g(r)} dr, \quad x \in \mathbb{R}.$$

Assume that h is twice continuously differentiable and that it has an inverse given by $h^{(-1)}$, in other words, there exists a function $h^{(-1)} : \mathbb{R} \rightarrow \mathbb{R}$ such that $h(h^{(-1)}(x)) = h^{(-1)}(h(x)) = x$ for any $x \in \mathbb{R}$.

Apply Itô's formula to $h(X_t)$ (make use of the Fundamental theorem of calculus) and use the invertibility of h to show that the solution X to the above stochastic differential equation can be written explicitly as

$$X_t = h^{(-1)}\left(W_t + h(x_0)\right), \quad \forall t \in [0, T].$$

[9 marks]

(turn page)

- (2) Suppose that $T \in (0, \infty)$, that $(W_t)_{t \in [0, T]}$ is a standard Brownian motion and, under the Black-Scholes market model assumption, that the price of a stock at time t is given by

$$S_t = S_0 e^{(r - \frac{\sigma^2}{2})t + \sigma W_t},$$

for $t \in [0, T]$, where $S_0 > 0$ is the initial price of the stock, $r > 0$ is the risk-free interest rate (continuous compounding) and $\sigma \in (0, \infty)$. Throughout Φ denotes the cumulative probability distribution function of the standard normal distribution.

- (a) Consider the *European parabola call option* with payoff

$$\Psi(S_T) := \max\{S_T^2 + 6S_T - 160, 0\}$$

at maturity time T in the standard Black-Scholes market model.

(i) Write the general expression from the *Fundamental Theorem of Asset Pricing* that yields the price $\Pi(0; \Psi(S_T))|_{S_0}$ of option $\Psi(S_T)$ at time $t = 0$. Give an interpretation.

(ii) Show that the price $\Pi(0; \Psi(S_T))|_{S_0}$ at $t = 0$ can be written as

$$\Pi(0; \Psi(S_T))|_{S_0} = e^{-rT} \left(\mathbb{E}[S_T^2 \mathbb{1}_{\{S_T > 10\}}] + 6\mathbb{E}[S_T \mathbb{1}_{\{S_T > 10\}}] - 160\mathbb{E}[\mathbb{1}_{\{S_T > 10\}}] \right),$$

where $\mathbb{1}_{\{S_T > 10\}}$ denotes the indicator function for the event $S_T > 10$. Briefly justify your arguments (with pictures if deemed necessary).

[7 marks]

- (b) We are offered the following European type option $\Psi(S_T)$ that allows us to get a reduction on the price S_T of the stock S when we wish to buy it at the expiry date T if S_T is higher than the spot price S_0 of the stock. Namely, depending on its value at maturity, we have the option to buy the stock at maturity for the following price

$$S_0 \text{ if } S_0 \leq S_T \leq 3S_0, \quad \text{and} \quad 2S_0 \text{ if } S_T > 3S_0, \quad (\text{zero otherwise})$$

(i) Sketch the graph for the contract's payoff function and write down an expression for the payoff at maturity T for a long position (as a single expression/function on S_T and other elements you may deem relevant).

(ii) Show that the fair price of the option at time $t = 0$ is

$$e^{-rT} S_0 \left(2 - \Phi(a) - \Phi\left(a + \frac{\ln 3}{\sigma \sqrt{T}}\right) \right), \quad \text{where} \quad a := \frac{\sqrt{T}}{\sigma} \left(\frac{1}{2} \sigma^2 - r \right).$$

[8 marks]

- (c) On our market two new contracts have appeared, the so-called *European Asian put* and *European Asian call* (with strike K , maturity time T and underlying S). The payoffs are given respectively by

$$\Psi_{\text{Asi call}} := \max \left\{ \frac{1}{T} \int_0^T S_t dt - K, 0 \right\} \text{ and } \Psi_{\text{Asi put}} := \max \left\{ K - \frac{1}{T} \int_0^T S_t dt, 0 \right\}.$$

Set $c_0 := \Pi(0; \Psi_{\text{Asi call}})$ and $p_0 := \Pi(0; \Psi_{\text{Asi put}})$ as the fair prices at time $t = 0$ of each option respectively.

(i) Use the explicit expression of $(S_t)_{t \in [0, T]}$ (and Fubini) to compute the quantity

$$-\frac{1}{T}e^{-rT}\mathbb{E}\left[\int_0^T S_t dt\right].$$

(ii) Show the following relation between the prices of the Asian Call and the Asian Put (on the same strike, underlying and maturity) holds

$$p_0 - c_0 = Ke^{-rT} + \frac{1}{rT}S_0(e^{-rT} - 1).$$

[10 marks]

